

Lecture-07 (17 August 2026)

Recap: A random variable X is said to be sub-Gaussian if any one of the following equivalent conditions are satisfied.

① There exists $K_1 > 0$ such that

$$P(|X| \geq \varepsilon) \leq 2 \exp\left(-\frac{\varepsilon^2}{K_1^2}\right) \quad \forall \varepsilon > 0.$$

② There exists $K_2 > 0$ such that

$$\left(E[|X|^p]\right)^{1/p} \leq K_2 \sqrt{p} \quad \forall p \geq 1.$$

③ There exists $K_3 > 0$ such that

$$E[e^{t^2 X^2}] \leq e^{t^2 K_3^2} \quad \forall t \in \left[-\frac{1}{K_3}, \frac{1}{K_3}\right].$$

④ There exists $K_4 > 0$ such that

$$E\left[e^{X^2/K_4^2}\right] \leq 2.$$

⑤ There exists $K_5 > 0$ such that

$$E\left[e^{t(X-E[X])}\right] \leq e^{t^2 K_5^2} \quad \forall t \in \mathbb{R}.$$

Definition (σ -sub-Gaussian)

Given $\sigma > 0$, a random variable X is said to be σ -sub-Gaussian if

$$E\left[e^{t(X-E[X])}\right] \leq e^{t^2 \sigma^2 / 2} \quad \forall t \in \mathbb{R}.$$

Properties: If X is σ -sub-Gaussian, then:

① $\text{Var}(X) \leq \sigma^2$.

② αX is $|\alpha|\sigma$ -sub-Gaussian for all $\alpha \in \mathbb{R}$.

③ If $X_1, \dots, X_n$ are IID copies of X , then $\frac{1}{n} \sum_{k=1}^n X_k$ is $\frac{\sigma}{\sqrt{n}}$ -sub-Gaussian.

④ If $E[X] = \mu$, then for all $\varepsilon > 0$,

$$P(X - \mu \geq \varepsilon) \leq \exp\left(-\frac{\varepsilon^2}{2\sigma^2}\right), \quad P(X - \mu \leq -\varepsilon) \leq \exp\left(-\frac{\varepsilon^2}{2\sigma^2}\right).$$

$$P\left(\frac{1}{n} \sum_{k=1}^n X_k - \mu \geq \varepsilon\right) \leq \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right),$$

$$P\left(\frac{1}{n} \sum_{k=1}^n X_k - \mu \leq -\varepsilon\right) \leq \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right).$$

$$\varepsilon = \sqrt{\frac{2\sigma^2}{n} \log \frac{2}{\delta}}$$

Setting $\delta = e^{-n\varepsilon^2/2\sigma^2}$, we get that

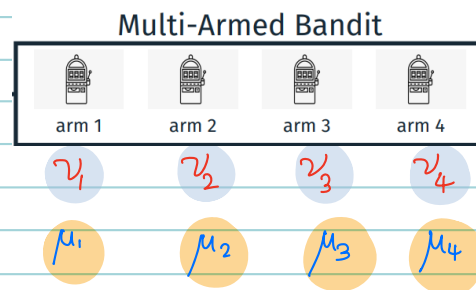
$$P\left(\left|\frac{1}{n} \sum_{k=1}^n X_k - \mu\right| \leq \sqrt{\frac{2\sigma^2}{n} \log \frac{2}{\delta}}\right) \geq 1 - \delta \quad \forall \delta > 0.$$

MULTI-ARMED BANDITS :

THE UPPER CONFIDENCE BOUND (UCB) ALGORITHM

- Consider a multi-armed bandit with K arms
- Let the distribution of rewards from arm k be $\mathcal{V}_k$, and let

$$E_{X \sim \mathcal{V}_k}[X] = \mu_k$$



- We assume that $(\mathcal{V}_1, \dots, \mathcal{V}_K)$ are **1-sub-Gaussian** and **unknown**
- Without loss of generality, let

$$\mu_1 > \max_{j \neq 1} \mu_j$$

(arm 1 is assumed to have the largest mean)

- At each time instant $t \in \{1, 2, \dots\}$, we are allowed to select one arm, say A_t , and observe its reward, say R_t $R_t \sim \mathcal{V}_{A_t}$
- The goal is to select the arms in such a way that we maximise

$$E\left[\sum_{t=1}^T R_t\right]$$

over a fixed, known time horizon of T time instants

Key optimisation problem : $\max E\left[\sum_{t=1}^T R_t\right]$

max is over all arm selection strategies

The performance of any arm selection strategy π is measured by its regret

$$R_T(\pi) := \underbrace{T\mu_1}_{\text{total mean reward possible if } \mu_1, \dots, \mu_K \text{ were known}} - \mathbb{E}^\pi \left[\sum_{t=1}^T R_t \right]$$

Example: Uniform arm selection strategy:

If $A_t \sim \text{unif}(\{1, \dots, K\})$ at every time instant, then

$$\begin{aligned} R_T(\pi_{\text{unif}}) &= T\mu_1 - \mathbb{E}^{\pi_{\text{unif}}} \left[\sum_{t=1}^T R_t \right] \\ &= T\mu_1 - \sum_{t=1}^T \mathbb{E}^{\pi_{\text{unif}}} [R_t] \\ &= T\mu_1 - \sum_{t=1}^T \mathbb{E}^{\pi_{\text{unif}}} \left[\mathbb{E}^{\pi_{\text{unif}}} [R_t | A_t] \right] \\ &= T\mu_1 - \sum_{t=1}^T \sum_{a=1}^K P^{\pi_{\text{unif}}}(A_t=a) \mathbb{E}^{\pi_{\text{unif}}} [R_t | A_t=a] \\ &= T\mu_1 - \sum_{t=1}^T \sum_{a=1}^K \frac{1}{K} \cdot \mu_a \\ &= T \left[\mu_1 - \frac{1}{K} \sum_{a=1}^K \mu_a \right] \end{aligned}$$

↑ regret grows linearly in $T = O(T)$!

Objective: To design an arm selection strategy whose regret grows sub-linearly with respect to T .

The UCB Algorithm

— Notations: For each arm $a \in \{1, \dots, K\}$, let:

- $N_a(t) := \#$ times arm a has been selected up to time t
Mathematically,

$$N_a(t) = \sum_{s=1}^t \mathbb{1}_{\{A_s=a\}}$$

- $\hat{\mu}_a(t) :=$ empirical mean of arm a as of time t

Mathematically,

$$\hat{\mu}_a(t) = \begin{cases} \frac{1}{N_a(t)} \sum_{s=1}^t \mathbb{1}_{\{A_s=a\}} R_s, & N_a(t) > 0 \\ 0, & N_a(t) = 0. \end{cases}$$

- $UCB_a(t, \delta) :=$ UCB index of arm a as of time t

Mathematically,

$$UCB_a(t, \delta) = \begin{cases} \hat{\mu}_a(t) + \sqrt{\frac{2 \log(1/\delta)}{N_a(t)}}, & N_a(t) > 0, \\ +\infty, & N_a(t) = 0. \end{cases}$$

Smaller the value of $N_a(t)$, larger the value of $UCB_a(t)$.

The UCB arm selection algorithm is as follows:

- Input $\delta \in (0, 1)$.
- Set $N_a(0) = 0 \quad \forall a$, $\hat{\mu}_a(0) = 0 \quad \forall a$
- For $t = 1, \dots, T$:
 - 1- Compute $\hat{\mu}_a(t-1)$ for each arm a
 - 2- Compute $UCB_a(t-1, \delta)$ for each arm a
 - 3- Select arm

$$A_t = \underset{a}{\operatorname{argmax}} UCB_a(t-1, \delta)$$

- 4- Observe reward R_t from arm A_t , and go to step 1

Theorem:

If $\delta = \frac{1}{T^2}$ in the UCB algorithm, then

$$R_T(\tau_{UCB}) \leq 3 \sum_{a=1}^k \Delta_a + \sum_{a: \Delta_a > 0} \frac{16 \log T}{\Delta_a},$$

Where $\Delta_a := \mu_1 - \mu_a$ for each $a \in \{1, \dots, k\}$
 $\rightarrow$ sub-optimality gap of arm a

Proof of Theorem:

It is a simple exercise to show that for any arm selection strategy π ,

$$R_T(\pi) = \sum_{a=1}^K \Delta_a \mathbb{E}^\pi [N_a(T)] = \sum_{a \neq 1} \Delta_a \mathbb{E}^\pi [N_a(T)].$$

We then have

$$R_T(\pi_{\text{UCB}}) = \sum_{a \neq 1} \Delta_a \mathbb{E}^{\pi_{\text{UCB}}} [N_a(T)].$$

Fix an arm $a \neq 1$, and define

$$g_a := \left\{ \min_{t \in \{1, \dots, T\}} \text{UCB}_1(t, \delta) > \mu_1, > \frac{1}{u_a} \sum_{t=1}^T 1_{\{A_t=a, N_a(t) \leq u_a\}} R_t + \sqrt{\frac{2}{u_a} \log \frac{1}{\delta}} \right\}$$

$\hookrightarrow$ a good event of interest to us. $\hookrightarrow u_a$ will be carefully designed in the proof

We will prove below that:

- ① If event g_a occurs under π_{UCB} , then $N_a(T) \leq u_a$.
- ② $\mathbb{P}^{\pi_{\text{UCB}}}(g_a) \geq 1 - \delta$. $\leftarrow g_a$ occurs under UCB with high probability

Assuming ① and ② are true, we have

$$\begin{aligned} \mathbb{E}^{\pi_{\text{UCB}}} [N_a(T)] &= \mathbb{E}^{\pi_{\text{UCB}}} \left[1_{g_a} N_a(T) \right] + \mathbb{E}^{\pi_{\text{UCB}}} \left[1_{g_a^c} N_a(T) \right] \\ &\leq u_a + T \mathbb{P}^{\pi_{\text{UCB}}}(g_a^c) \rightarrow \textcircled{a} \end{aligned}$$

Proof of ①: Suppose that g_a occurs. Suppose further that $N_a(T) > u_a$.

- This means that arm a was selected more than u_a times up to time T . This in turn means that there must have been a time instant $t^* \in \{1, \dots, T-1\}$ such that

$$N_a(t^*-1) = u_a, \quad A_{t^*} = a.$$

- We then have

$$\begin{aligned} \text{UCB}_a(t^*-1, \delta) &= \hat{\mu}_a(t^*-1) + \sqrt{\frac{2 \log V_\delta}{N_a(t^*-1)}} \\ &= \frac{1}{N_a(t^*-1)} \sum_{t=1}^{t^*-1} 1_{\{A_t=a\}} R_t + \sqrt{\frac{2 \log(V_\delta)}{N_a(t^*-1)}} \end{aligned}$$

$$N_a(t^*-1) = u_a \leq \frac{1}{u_a} \sum_{t=1}^T \frac{1}{\{A_t=a, N_a(t) \leq u_a\}} R_t + \sqrt{\frac{2 \log(V_8)}{u_a}} < \mu_1 < \text{UCB}_1(t^*-1, \delta) \quad (\text{by the def}^n \text{ of } \mathcal{G}_a),$$

which means that

$$A_{t^*} = \arg\max_a \text{UCB}_a(t^*-1, \delta) \neq a \quad (\text{contradiction})$$

Thus, we must have

$$\mathcal{G}_a \subseteq \{N_a(T) \leq u_a\}.$$

Proof of ②:

Notice that

$$\mathcal{G}_a^c = \underbrace{\left(\bigcup_{t=1}^T \{ \text{UCB}_1(t, \delta) \leq \mu_1 \} \right)}_{\mathcal{G}_{a,1}^c} \cup \underbrace{\left\{ \frac{1}{u_a} \sum_{t=1}^T \frac{1}{\{A_t=a, N_a(t) \leq u_a\}} R_t + \sqrt{\frac{2 \log(V_8)}{u_a}} \geq \mu_1 \right\}}_{\mathcal{G}_{a,2}^c}$$

We then have

$$\begin{aligned} \mathbb{P}^{\pi_{\text{UCB}}}(\mathcal{G}_{a,1}^c) &= \mathbb{P}^{\pi_{\text{UCB}}} \left(\bigcup_{t=1}^T \{ \text{UCB}_1(t, \delta) \leq \mu_1 \} \right) \\ &\leq \sum_{t=1}^T \mathbb{P}^{\pi_{\text{UCB}}} \left(\text{UCB}_1(t, \delta) \leq \mu_1 \right) \quad (\text{union bound}) \\ &= \sum_{t=1}^T \mathbb{P}^{\pi_{\text{UCB}}} \left(\hat{\mu}_1(t) + \sqrt{\frac{2 \log(V_8)}{N_1(t)}} \leq \mu_1 \right) \\ &= \sum_{t=1}^T \mathbb{P}^{\pi_{\text{UCB}}} \left(\frac{1}{N_1(t)} \sum_{s=1}^t \frac{1}{\{A_s=1\}} R_s - \mu_1 \leq -\sqrt{\frac{2 \log(V_8)}{N_1(t)}} \right) \\ &\leq \sum_{t=1}^T \mathbb{P}^{\pi_{\text{UCB}}} \left(\underbrace{\frac{1}{N_1(t-1)} \sum_{s=1}^t \frac{1}{\{A_s=1\}} R_s}_{\text{Sum of IID rewards from arm 1 up to time } t} - \mu_1 \leq -\underbrace{\sqrt{\frac{2 \log(V_8)}{T}}}_{\varepsilon} \right) \end{aligned}$$

We have (using sub-Gaussian concentration)

$$\mathbb{P}^{\pi_{\text{UCB}}} \left(g_{a,1}^c \right) \leq \sum_{t=1}^T \exp \left(- \frac{T \epsilon^2}{2} \right) \quad (\nu_1 \text{ is 1-sub-Gaussian})$$

$$= T \delta. \quad \rightarrow \textcircled{a1}$$

To upper bound $\mathbb{P}^{\pi_{\text{UCB}}} \left(g_{a,2}^c \right)$, we choose u_a such that

$$\Delta_a - \sqrt{\frac{2 \log(V_8)}{u_a}} \geq c \Delta_a, \quad \rightarrow \textcircled{a1-1}$$

where $c \in (0,1)$ will be chosen later. Then, we have

$$\mathbb{P}^{\pi_{\text{UCB}}} \left(g_{a,2}^c \right) = \mathbb{P}^{\pi_{\text{UCB}}} \left(\frac{1}{u_a} \sum_{t=1}^T \mathbb{1}_{\{A_t=a, N_a(t) \leq u_a\}} R_t + \sqrt{\frac{2 \log(V_8)}{u_a}} \geq \mu_1 \right)$$

$$= \mathbb{P}^{\pi_{\text{UCB}}} \left(\frac{1}{u_a} \sum_{t=1}^T \mathbb{1}_{\{A_t=a, N_a(t) \leq u_a\}} R_t - \mu_a \geq \Delta_a - \sqrt{\frac{2 \log(V_8)}{u_a}} \right)$$

$$\leq \mathbb{P}^{\pi_{\text{UCB}}} \left(\frac{1}{u_a} \sum_{t=1}^T \mathbb{1}_{\{A_t=a, N_a(t) \leq u_a\}} R_t - \mu_a \geq \underbrace{c \Delta_a}_{\epsilon} \right)$$

$$\leq \exp \left(- \frac{u_a \epsilon^2}{2} \right) \quad (\nu_a \text{ is 1-sub-Gaussian})$$

$$= \exp \left(- u_a \frac{c^2 \Delta_a^2}{2} \right). \quad \rightarrow \textcircled{a2}$$

Putting back $\textcircled{a1}$ and $\textcircled{a2}$ into $\textcircled{a}$, we get

$$\mathbb{E}^{\pi_{\text{UCB}}} [N_a(T)] \leq u_a + T \left(T \delta + \exp \left(- u_a \frac{c^2 \Delta_a^2}{2} \right) \right).$$

$$= u_a + T^2 \delta + T \exp \left(- u_a \frac{c^2 \Delta_a^2}{2} \right).$$

We choose u_a to be the smallest integer such that $\textcircled{a1-1}$ holds:

$$u_a = \left\lceil \frac{2 \log(V_8)}{(1-c)^2 \Delta_a^2} \right\rceil.$$

With this choice of u_a , setting $\delta = \frac{1}{T^2}$, we get

$$E^{\pi_{UCB}} [N_a(T)] \leq \left\lceil \frac{2 \log T^2}{(1-c)^2 \Delta_a^2} \right\rceil + 1 + T^{1 - \frac{2c^2}{(1-c)^2}}.$$

Setting $c = \frac{1}{2}$, we get

$$\begin{aligned} E^{\pi_{UCB}} [N_a(T)] &\leq \left\lceil \frac{16 \log T}{\Delta_a^2} \right\rceil + 1 + \frac{1}{T} \\ &\leq 3 + \frac{16 \log T}{\Delta_a^2} \quad \forall a \neq 1. \end{aligned}$$

Finally, we have

$$\begin{aligned} R_T(\pi^{UCB}) &= \sum_{a \neq 1} \Delta_a E^{\pi_{UCB}} [N_a(T)] \\ &\leq \sum_{a \neq 1} \Delta_a \left(3 + \frac{16 \log T}{\Delta_a^2} \right) \\ &\leq 3 \sum_{a=1}^K \Delta_a + \sum_{a \neq 1} \frac{16 \log T}{\Delta_a}. \end{aligned}$$

Remarks:

① The regret of UCB algorithm grows like

$$R_T(\pi_{UCB}) = O(\log T).$$

The natural question is: can we do better than $O(\log T)$?

The answer is NO! It turns out that

$$R_T(\pi) = \Omega(\log T) \quad \forall \pi.$$

② The regret bound we have computed for UCB is instance-dependent
depends on $\mu_1, \dots, \mu_K$